Kinetrics Canada — AI Engineer Intern Experience

Project Description

At Kinetrics Canada, I worked on generating realistic synthetic images of nuclear fuel bundles to solve the problem of data scarcity. Because nuclear assets are located in sensitive environments and subject to strict security protocols, capturing actual images is extremely difficult. By fine-tuning a Stable Diffusion model on our manually collected dataset, I created a pipeline that generates clear, diverse images of various fuel bundle shapes (cylindrical, rectangular, and octagonal). This synthetic data helps bypass physical access barriers and provides a secure, high-quality alternative dataset for engineering and testing.

Key Responsibilities

- Cleaned, resized, and categorized the dataset of 307 real-world images into shape-specific groups, then created prompt descriptions and metadata files to guide model training.
- Fine-tuned CompVis Stable Diffusion v1.4 using PyTorch and Hugging Face libraries, adjusting the model to recognize distinct shape features.
- Used optimization tools like 8-bit Adam (bitsandbytes) and mixed-precision (fp16 via Accelerate) to manage memory and train efficiently on Google Colab T4 GPUs.
- Conducted training runs across different checkpoints (1,000 to 5,000 epochs) to find the sweet spot, stopping at 3,000 epochs to prevent overtraining and loss of image diversity.
- Evaluated the quality and diversity of the generated images by plotting Fréchet Inception Distance (FID) and Inception Score (IS) metrics.
- Collaborated on final documentation and technical reports, demonstrating how the best-performing model matched real-world fuel bundle dimensions and features.

Technical Environment

Python, PyTorch, Hugging Face (Diffusers, Transformers, Datasets), Accelerate, Bitsandbytes (8-bit Adam), Google Colab, Linux, Torchmetrics (FID/IS evaluation), Torch-fidelity, Matplotlib, Git.